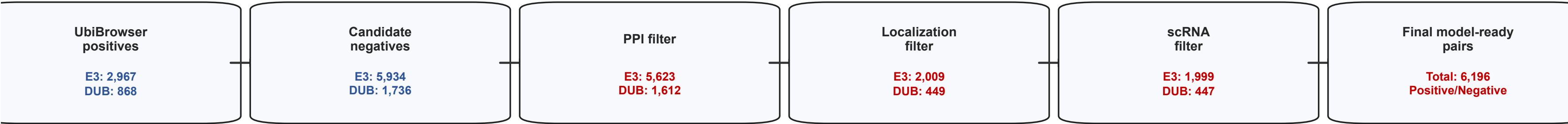
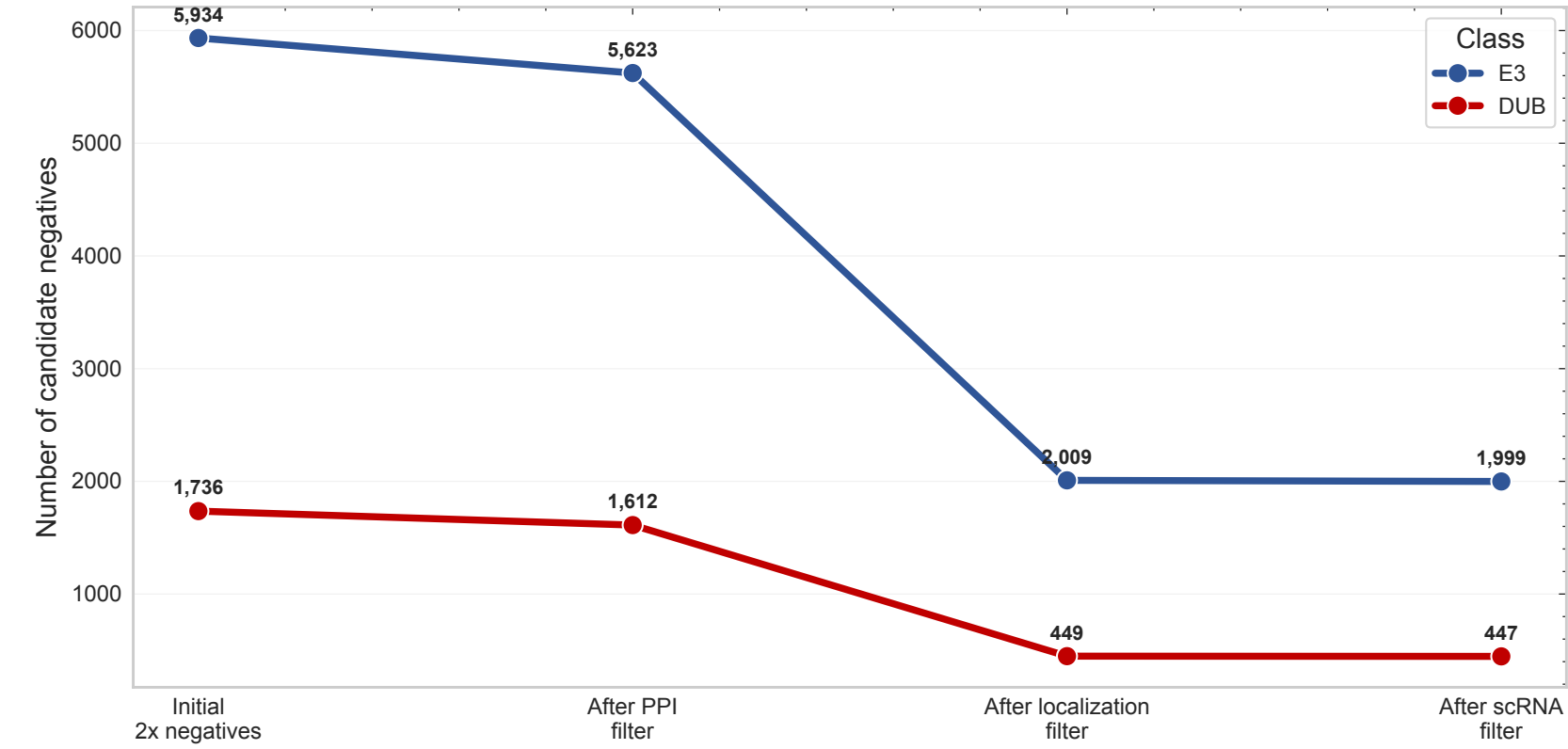


Figure 3. Construction and quality control of the E3/DUB interaction dataset

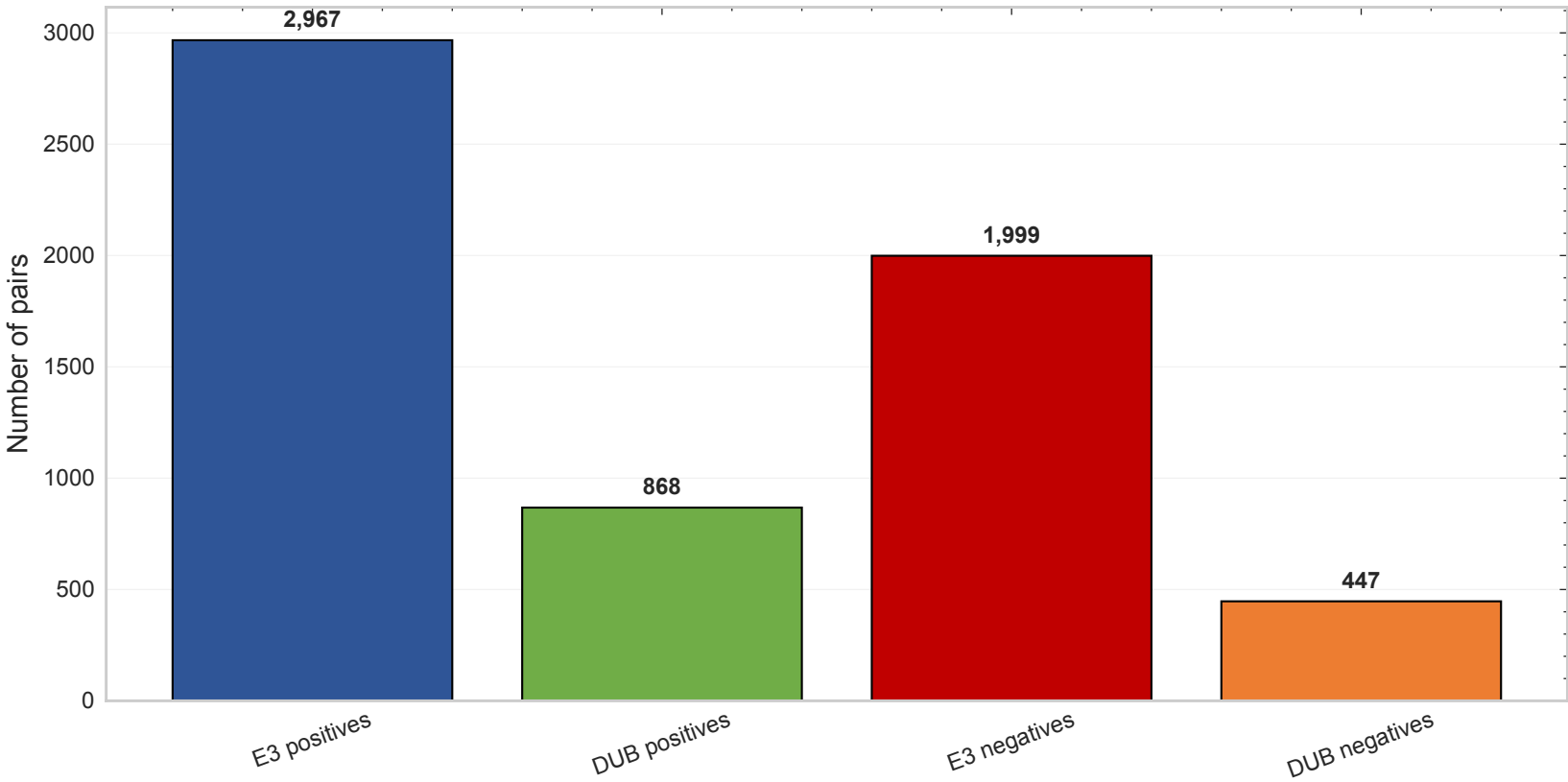
A. Dataset construction and context-aware negative filtering



B. Negative set reduction across biological filters



C. Final class composition



D. Final dataset QC summary

QC item	Value
Final labeled pairs	6,196
Unique pair IDs	6,196
Positive labels	3,806
Negative labels	2,390
Duplicate pair IDs	0
Unique enzymes	497
Unique substrates	1,788